

Exploring Chemistry

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![[Cover]](h7fig:cover)

****Atoms through organic chemistry**** — periodic table, reactions, stoichiometry.

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1. Atomic structure

- Proton, neutron, electron — atomic number Z , mass A
- Isotopes — same Z , different neutron count
- Electron configuration — shells, orbitals, Aufbau

2. Periodic table regions

Region	Traits	Examples
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Alkali metals	Soft, +1 ion	Li, Na, K
Halogens	Reactive nonmetals	F, Cl, Br
Noble gases	Inert full shell	He, Ne, Ar
Transition metals	Variable oxidation	Fe, Cu, Zn
Lanthanides/actinides	f-block	U, Pu

3. Bonding & reactions

- Ionic, covalent, metallic, hydrogen bonds
- Balancing equations — conservation of mass
- Acid-base — pH, neutralization
- Redox — oxidation states, electron transfer
- Thermochemistry — enthalpy, exo vs endo

4. Organic overview

- Hydrocarbons — alkane, alkene, alkyne, aromatic
- Functional groups — alcohol, carbonyl, carboxyl, amine
- Polymers — addition vs condensation

5. AI retrieval keys

Index: `element`, `bond`, `reaction\_type`, `functional\_group`. Dewey **540**. Pair *Exploring Biology*.